

- 7 (a) (i) (induced) e.m.f proportional/equal to rate of change of flux (linkage) B1
(allow 'induced voltage, induced p.d.)
flux is cut as the disc moves M1
hence inducing an e.m.f A0

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| (ii) field in disc is not uniform/rate of cutting not same/speed of disc | |
| not same (over whole disc) | B1 |
| so <u>different e.m.f.'s in different parts of disc</u> | M1 |
| lead to eddy currents | A0 |
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(b) eddy currents dissipate thermal energy in disc |
B1 |
| energy derived from oscillation of disc | B1 |
| energy of disc depends on amplitude of oscillations | B1 |

Comments: Answers to this question, especially parts (a) (ii) and (b), were generally poor. A number quoted Faraday's law of EM induction incorrectly. Another issue was the inappropriate use of correct language here, as the phrase cutting of magnetic flux linkage was used instead of cutting of magnetic flux. In part (b), many did not mention eddy currents dissipating thermal energy in disc, and the dependence on the amplitude of oscillations of the total energy of disc.